MarkMate

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PROJECT TITLE

Student Mark System

STUDENT DETAILS

Names: Kiseki Gilili, Hlonela Sompondo, Banoyolo Ngeva, Siphosethu Ngxwana, Luniko Funde

INTRODUCTION

This System Analysis Document (SAD) follows on from the Capstone Project Proposal for the Student Mark System. It explains how student marks are currently managed, identifies the problems with the current system, and describes what the new system must do. This analysis will guide the design and development of the project.

ANALYSIS OF THE CURRENT SYSTEM (AS-IS)

4. 1) CURRENT PROCESS

At present, teachers record student marks manually using paper records or spreadsheets. Marks for tests, assignments and exams are entered individually and final marks are calculated by hand. Finding previous records and preparing reports can take a lot of time, especially when dealing with many students.

4. 2) PROBLEMS WITH CURRENT SYSTEM

1. Recording and updating marks manually takes a lot of time.
2. Human errors can occur when entering or calculating marks.
3. Student records can be lost, damaged or misplaced.
4. Searching for a student's marks is slow.
5. Preparing reports and final results is time consuming.

STAKEHOLDERS (USERS OF THE SYSTEM)

STAKEHOLDER	ROLE	WHAT THEY NEED FROM THE SYSTEM
Teacher/ Lecturer	Records and updates student marks	A quick and accurate way to capture, edit and calculate marks
Administrator	Manages student records and system data	Secure access to manage and users and student information
Student	Recieves marks and results	Accurate and up-to-date academic records

REQUIREMENTS ANALYSIS

FUNCTIONAL REQUIREMENTS

1. The system must allow teachers to log in securely.
2.The system must allow teachers to add new student records.
3.The system must allow teachers to capture marks for tests, assignments and exams.
4.The system must automatically calculate final marks.
5.The system must allow teachers to edit or update marks.
6. The system must store all student information in a MySQL database.
7.The system must allow teachers to search for student records.
8.The system must generate reports showing student performance.

NON-FUNCTIONAL REQUIREMENTS

1.The system should load pages within three seconds under normal conditions.
2.The system should be simple and easy to use.
3.Only authorised users should have access to student records.
4.Student data must be stored securely without loss.
5.The system should provide accurate calculations every time.

PROCESS ANALYSIS

MAIN SYSTEM PROCESSES

Login: The teacher enters a username and password to access the system.
Select Student: The teacher searches for or selects a student
Capture Marks: Marks for tests, assignments or exams are entered
Calculate Final Mark: The system automatically calculates the student's final mark
Save Record: The information is stored in the MySQL database
Generate Report: The teacher views or prints student performance reports.

FEASIBILITY STUDY

TECHNICAL FEASIBILITY

The project uses Python (Django), HTML, MySQL and Visual Studio Code, which are suitable for building a web-based Student Mark System. These tools are reliable, widely used and achievable within the team's current skill level.

OPERATIONAL FEASIBILITY

Teachers already use computers to manage student information, so learning the new system should not be difficult. The system will improve the current process by making it faster and more accurate.

ECONOMIC FEASIBILITY

The software required for the project is free or open-source, so no additional costs are expected apart from existing computer and internet resources.

SCHEDULE FEASIBILITY

The 5 week project schedule provides enough time to complete the planning, design, development, testing and presentation of the Student Mark System.

SYSTEM REQUIREMENTS SPECIFICATION

HARDWARE REQUIREMENTS

-A laptop or desktop computer with at least 4 GB RAM.
-Stable internet connection for research and testing.

SOFTWARE REQUIREMENTS

-Python programming language
-Django framework
-MySQL database
-HTML
-Visual Studio Code
-Modern web browser (Google Chrome, Microsoft Edge)

ASSUMPTIONS AND CONSTRAINTS

ASSUMPTIONS

-Teachers have access to a computer with internet access.
-Student information is already available for capturing into the system.
-Users have basic computer skills.

CONSTRAINTS

-The system will only manage student marks and related reports.
-The project must be completed within the five-week timeline.
-Only authorised users can access or edit student records.

CONCLUSION

This System Analysis Document has identified the weaknesses of the current manual method of managing student marks and outlined the requirements for the new Student Mark System. It explains the users of the system, the data that will be stored, and the functional and non functional requirements. This analysis provides a solid foundation for the design, development and implementation of the Student Mark System.